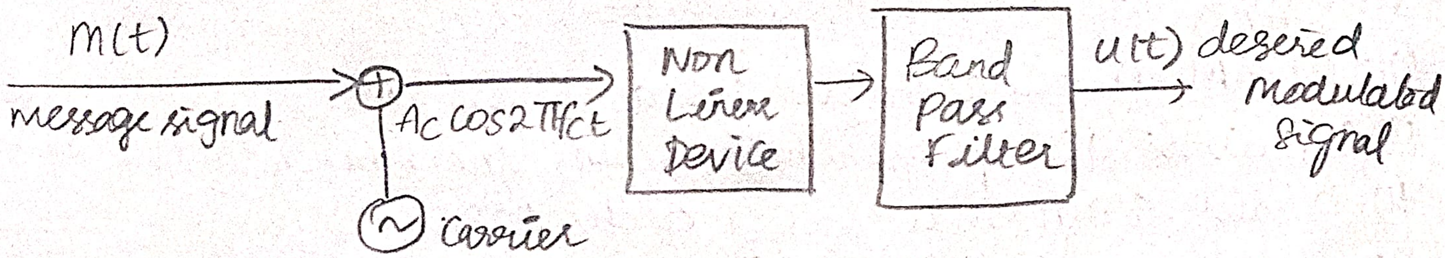
Modulation

1) AM - Square law Modulator

Square law diode modulation makes use of non linear current voltage charatteristics of diode.

Principle of square law

When two different frequencies are simultaneously passed through a non linear device, such as diode the process of amplitude modulation occurs.



The resulting voltage will be given by square law equation,

$$V_0(t) = a_1 V_i(t) + a_2 V_i^2(t) \rightarrow ①$$

$$V_i(t) = m(t) + A_c \cos 2\pi f_c t \rightarrow ②$$

Subs ② in ①

$$V_0(t) = a_1[m(t) + A_c \cos 2\pi f_c t] + a_2[m(t) + A_c \cos 2\pi f_c t]$$

$$= a_1 m_1(t) + a_1 A_c \cos 2\pi f_c t + a_2[m^2(t) + A_c^2 \cos^2 2\pi f_c^2 t + 2m(t) A_c \cos 2\pi f_c t)$$

$$V_0(t) = a_i V_i(t) + a_2 V_i^2(t)$$

$$V_i(t) = V_m(t) + V_c(t) = V_m \sin \omega_m t + V_c \sin \omega_c t$$

$$V_0(t) = a_1(V_m \sin \omega_m t + V_c \sin \omega_c t) + a_2(V_m \sin \omega_m t + V_c \sin \omega_c t)^2$$

$$= a_1 V_m \sin \omega_m t + a_1 V_c \sin \omega_c t + a_2 V_m^2 \sin^2 \omega t + a_2 V_c^2 \sin^2 \omega_c t + 2a_2 V_m V_c \sin \omega_m t \sin \omega_c t$$

$$= a_1 V_m \sin \omega_m t + a_1 V_c \sin \omega_c t + 2 a_2 V_m V_c \sin \omega_m t \sin \omega_c t$$

$= a_1 V_m \sin\omega_m t + a_1 V_c \sin\omega_c t + \frac{\cancel{2} a_2 V_m V_c}{\cancel{2}} [\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t]$

$V_o(t) = a_1 V_c \sin\omega_c t + a_2 V_m V_c [\cos(\omega_c - \omega_m)t - \cos(\omega_c + V_m)t]$

Hence AM wave is generator.

2) <u>DSB-SC</u> → <u>Balanced Modulation</u>

<u>Principle</u>: It cancels the carrier using phase cancellation.

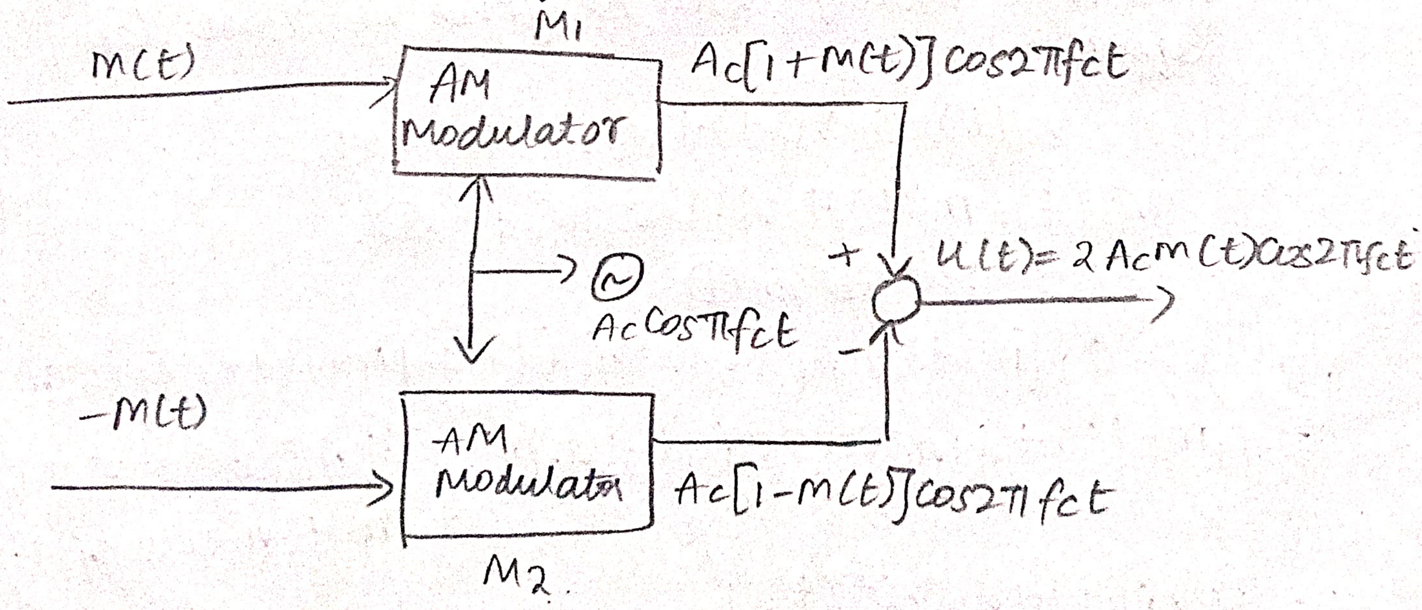
<u>Working</u>

#) Two AM modulators are used

#) Carrier is 180° phase shifted in one branch

#) Carrier components cancel each other

#) Sidebands add together



O/P of M1 $S_1(t) = A_c [1 + m(t)] \cos 2\pi f_c t$

O/P of M2 $S_2(t) = A_c [1 - m(t)] \cos 2\pi f_c t$

$u(t) = S_1(t) - S_2(t)$

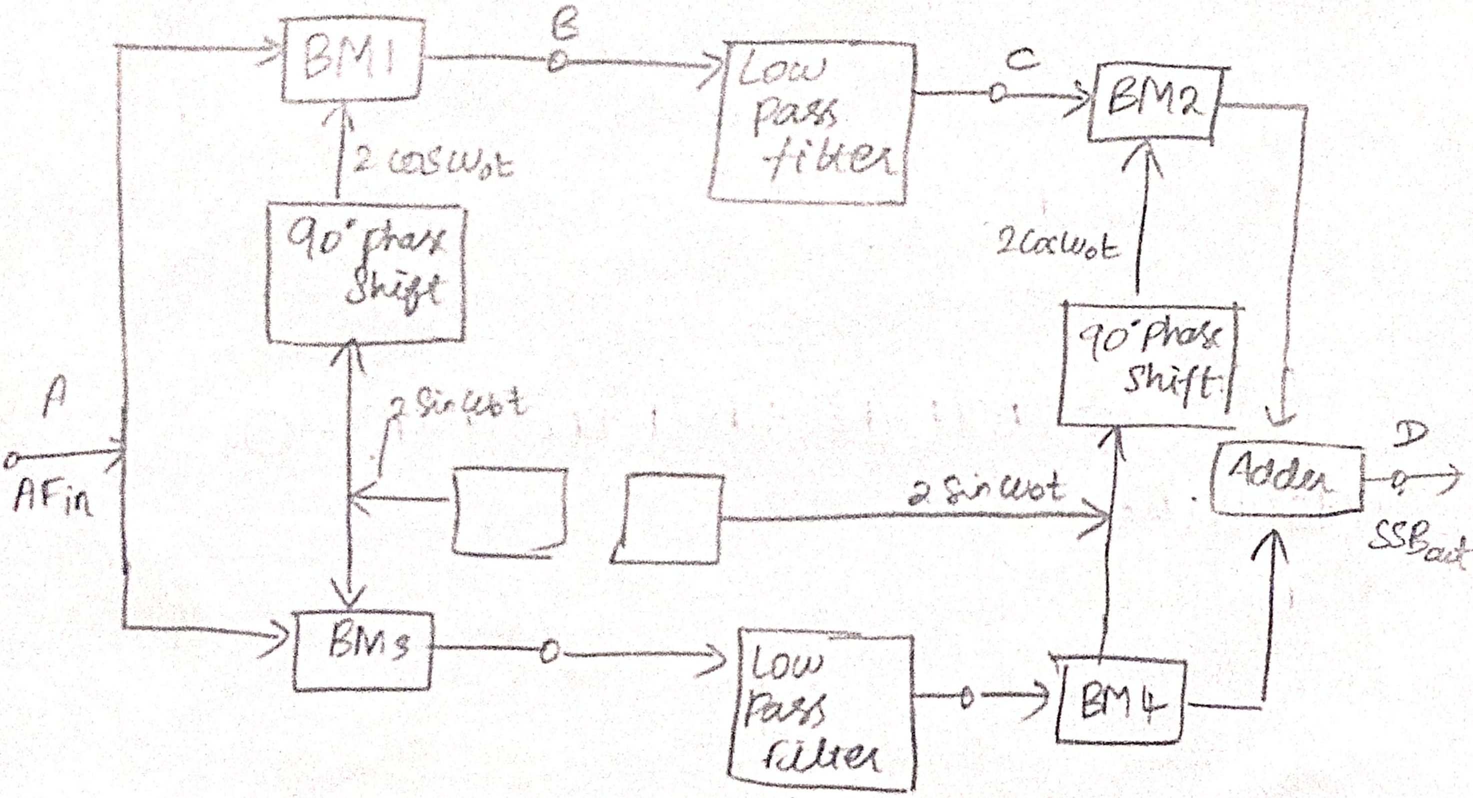
$= A_c [1 + m(t)] \cos 2\pi f_c t - A_c [1 - m(t)] \cos 2\pi f_c t$

$= A_c \cos \cancel{2}\pi f_c t + A_c m(t) \cos 2\pi f_c t - A_c \cos \cancel{2}\pi f_c t + A_c m(t) \cos 2\pi f_c t$

$\boxed{u(t) = 2 A_c m(t) \cos 2\pi f_c t}$

3) SSB-Weaver's method.

It is an improved version of the phase shift method used to generate SSB-SC signal. It reduces practical difficulties of generating accurate 90° phase shift at higher frequencies.



Steps

1) multiply signal

1st modulation o/p : $m(t)\cos(2\pi f_c t)$

2nd modulation o/p : $\hat{m}(t)\sin(2\pi f_c t)$

2) combine o/p

For USB, $s(t) = m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t)$

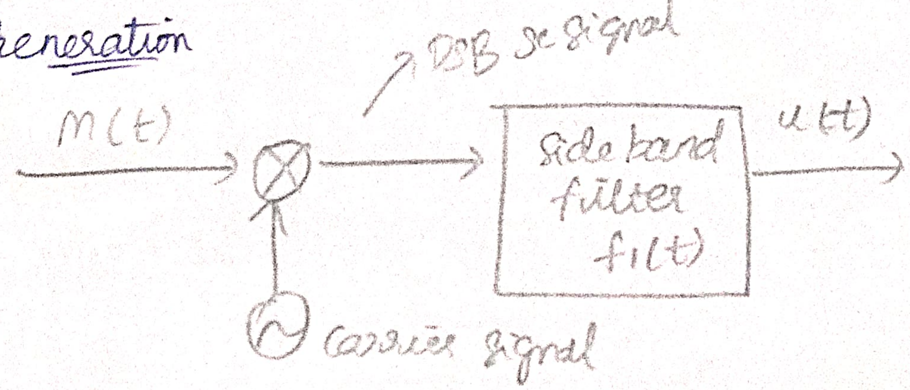
For LSB, $s(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t)$

One sideband cancels due to phase relationship.

4) VSB - one generator

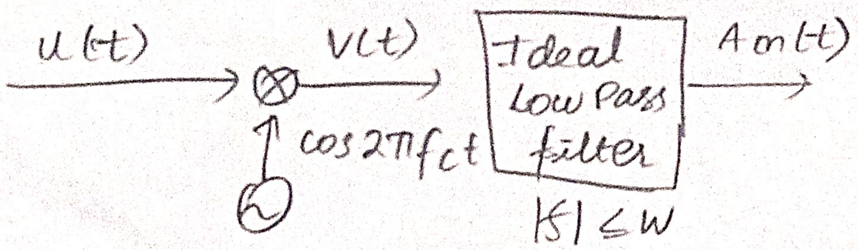
One of the side band is partially suppressed and vestigial of the other sideband is transmitted. The vestige compensates the suppression of sideband. It is called vestigial sideband.

Generation



$$u(F) = \frac{A_c}{2}[m(F-F_c) + M(F+F_c)]H(F) \rightarrow ①$$

Demodulation



$$V(t) = u(t)\cos 2\pi f_c t$$

$$V(f) = \frac{1}{2}[u(f-f_c) + U(f+f_c)] \rightarrow ②$$

Subs ① in ②

$$① \Rightarrow U(F-F_c) = \frac{A_c}{2}[m(F-F_c) + M(F-F_c+F_c)] + H(F-F_c)$$

$$= \frac{A_c}{2}[m(F-2F_c) + M(F)]H(F-F_c)$$

$$U(F+F_c) = \frac{A_c}{2}[m(F+F_c-F_c) + M(F+F_c+F_c)]H(F+F_c)$$

$$= \frac{A_c}{2}[M(F) + M(F+2F_c)]H(F+F_c)$$

$$V(f) = \frac{1}{2}\left[\frac{A_c}{2}[m(F-2F_c) + M(F)]H(F-F_c)] + \frac{A_c}{2}[m(F) + M(F+2F_c)]H(F+F_c)\right]$$

$$= \frac{A_c}{4}\left[M(F-\cancel{2F_c})+M(F)\right]H(F-F_c)+\left[M(F)+M(F+\cancel{2F_c})\right]H(F+\cancel{f_c})$$

$$= \frac{A_c}{4} M(F)\left[H(F-F_c)+H(F+f_c)\right]$$

VSB filter must satisfy the condition

$$H(F-F_c)+H(f+f_c) = \text{constant}$$

$$|f| \le W$$

<u>Bandwidth</u> : $BW = F_{USB} - F_{LSB}$

$$= F_c + F_m - (F_c - F_v)$$

$$= \cancel{F_c} + F_m - \cancel{F_c} + F_v$$

$$\boxed{BW = F_v + W}$$